

ArtData™ Module

AD-I — ArtData™ Integrity Module

ArtData™ Standard System

Module ID: AD-I

Standard System: ArtData™

Category: AI & Data Integrity Standards (AI)

Subcategory: Dataset Integrity

Version: 1.0

Status: Canonical · Module

Compatibility: ArtData™ Standard · MTVF™ · AI Governance Frameworks

Canonical Language: English

About the Module

What the Integrity Module Is

The **ArtData™ Integrity Module (AD-I)** defines the structural conditions required to preserve the integrity of datasets used in artificial intelligence systems.

AI models depend on datasets that remain stable, verifiable, and resistant to unauthorized modification.

Without structural integrity controls, datasets may change unnoticed, leading to unreliable training environments and non-reproducible AI behavior.

The Integrity Module introduces minimum mechanisms that allow dataset state to be verified, monitored, and protected against silent modification.

What This Module Changes

In many AI environments datasets are:

- modified without version control
- partially overwritten during preprocessing
- altered without recorded verification

The Integrity Module introduces a structural rule:

every dataset must have a verifiable integrity reference.

This allows organizations to detect dataset modification and maintain confidence in AI training data.

Canonical Definition

ArtData™ Integrity is the structural ability to verify that a dataset has not been altered without documentation, using identity anchoring, integrity references, and recorded modification history.

Dataset integrity ensures that AI systems operate on stable and verifiable data foundations.

Scope

The Integrity Module applies to datasets used in:

- AI training systems
- machine learning evaluation datasets
- model validation environments
- simulation and testing datasets
- automated decision system training pipelines

The module focuses on dataset immutability and modification verification.

Structural Requirements

To satisfy ArtData™ Integrity conditions, the following must be established.

Dataset Integrity Reference

Each dataset must possess an integrity verification mechanism.

Examples include:

- cryptographic hash
 - checksum reference
 - dataset signature
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Dataset Version Control

Datasets must maintain version continuity.

Minimum requirement:

- dataset version identifier
 - version history record
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Modification Detection

Any change to dataset content must be detectable.

Examples:

- new dataset version created
 - integrity reference updated
 - modification log recorded
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Integrity Verification Record

Integrity verification events must be documented.

Examples:

- integrity verification date
 - verification method used
 - verification result
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Minimum Implementation Framework (MIF)

Implementation Steps

Step 1 — Generate Dataset Integrity Reference

Create an integrity reference for the dataset.

Examples:

- SHA-256 hash
 - checksum reference
 - dataset fingerprint
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Step 2 — Establish Dataset Versioning

Assign version identifiers to the dataset.

Example:

- Dataset Version 1.0
- Dataset Version 1.1
- Dataset Version 2.0

Each new modification requires a new version reference.

Step 3 — Record Integrity Verification

Maintain a record confirming dataset integrity.

Minimum information:

- verification method
 - verification timestamp
 - dataset version verified
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Step 4 — Document Dataset Modifications

When dataset changes occur:

- modification must be recorded
 - dataset version updated
 - new integrity reference generated
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Architecture Position

The Integrity Module represents the dataset stability layer within the ArtData™ architecture.

ARTDATA™ DATA INTEGRITY STRUCTURE

Dataset Origin (AD-P)
↓
Dataset Integrity (AD-I)
↓
Lifecycle Continuity
↓
Transformation Transparency
↓
Responsible Entity

Integrity ensures that datasets remain stable and verifiable after their origin is established.

Use Case 1

Stable AI Training Environment

An AI company trains models using a dataset collected from multiple sources.

Without integrity verification, the dataset may change during preprocessing or system migration.

By implementing the Integrity Module:

- dataset hash is generated
- dataset versioning is implemented
- integrity verification records are maintained

Result:

- dataset stability is preserved
 - training environments remain reproducible
 - model development becomes more reliable
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Use Case 2

AI Model Audit and Dataset Verification

An organization must verify whether a dataset used in model training has been altered since initial deployment.

Using the Integrity Module:

- dataset integrity reference is checked
- dataset version history is reviewed
- modification logs are analyzed

Result:

- dataset authenticity can be verified
 - unauthorized modification can be detected
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- AI model audit becomes possible
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Canonical Closing Statement

Reliable AI requires datasets that remain verifiable over time.

The ArtData™ Integrity Module establishes the structural conditions required to preserve dataset integrity.

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Canonical Source

<https://originopenfoundation.org/>